

AI-Driven Tools for Medical Imaging and Visualization

Winter Semester 26/27

Kick-off Presentation

July 2nd, 2026

Who are we?



Computational Imaging and AI in Medicine
Prof. Dr. Julia Schnabel

Who are we?

Fryderyk Kögl



Research Interests:

- Deep Learning-Based Image Registration
- Data Curation & Visualization
- Neuronavigation

fryderyk.koegl@tum.de

Dr. Alejandra Castelblanco



Research Interests:

- VR for Medical Imaging
- Remote Health Monitoring
- Biomedical Signal Processing
- Digital Health Technologies

alejandra.castelblanco@tum.de

Dr. Cristina González



Research Interests:

- Multi-modal Learning
- Medical Image Segmentation
- Foundation Models

ci.gonzalez@tum.de

Seminar's Structure

Week	Session
1	Introduction to the seminar
2	How to read a paper and do poster presentations
3	Medical image registration
4	Invited talk
5	Student presentations
6	Student presentations
7	Medical image segmentation
8	Invited talk
9	Student presentations
10	Student presentations
11	Virtual and augmented reality
	Christmas Break
12	Invited talk
13	Student presentations
14	Student presentations
15	Poster presentations

Deliverables & Grading

Oral presentation:

- 30 min presentations – 10 min questions
- Presentation date depends on the topic

Poster presentation:

- 3 – 5 min pitch
- End of the semester

Paper selection:

- We provide a list of papers
- Assignment of papers for presentations after the seminar (moodle)
- Papers on both CV methodologies or adaptations to medical images

Grading:

- We will take into account the following things:
 - Oral presentation
 - Poster presentation
 - Attendance
 - Participation

Seminar's Goals

Theoretical knowledge:

Understand cutting-edge techniques and their direct applications in clinical practice.

(i) Clinical Visualization & Navigation

- Traditional medical imaging workflows (e.g., MRI, ultrasound and endoscopy).

(ii) Medical Image Registration

- Cross-modality registration (combining different scan types).
- Temporal registration (tracking changes over time).

(iii) Medical Image Segmentation

- Fully automated deep learning models.
- Interactive, promptable segmentation for precision targeting.

(iv) Virtual and Augmented Reality

- High-fidelity 3D rendering (e.g., Gaussian Splatting).
- Virtual and Augmented Reality (VR/AR) integrations in the clinic.

Research skills:

- How to read and present a scientific paper.
- How to design and present a scientific poster.

Medical Image Registration

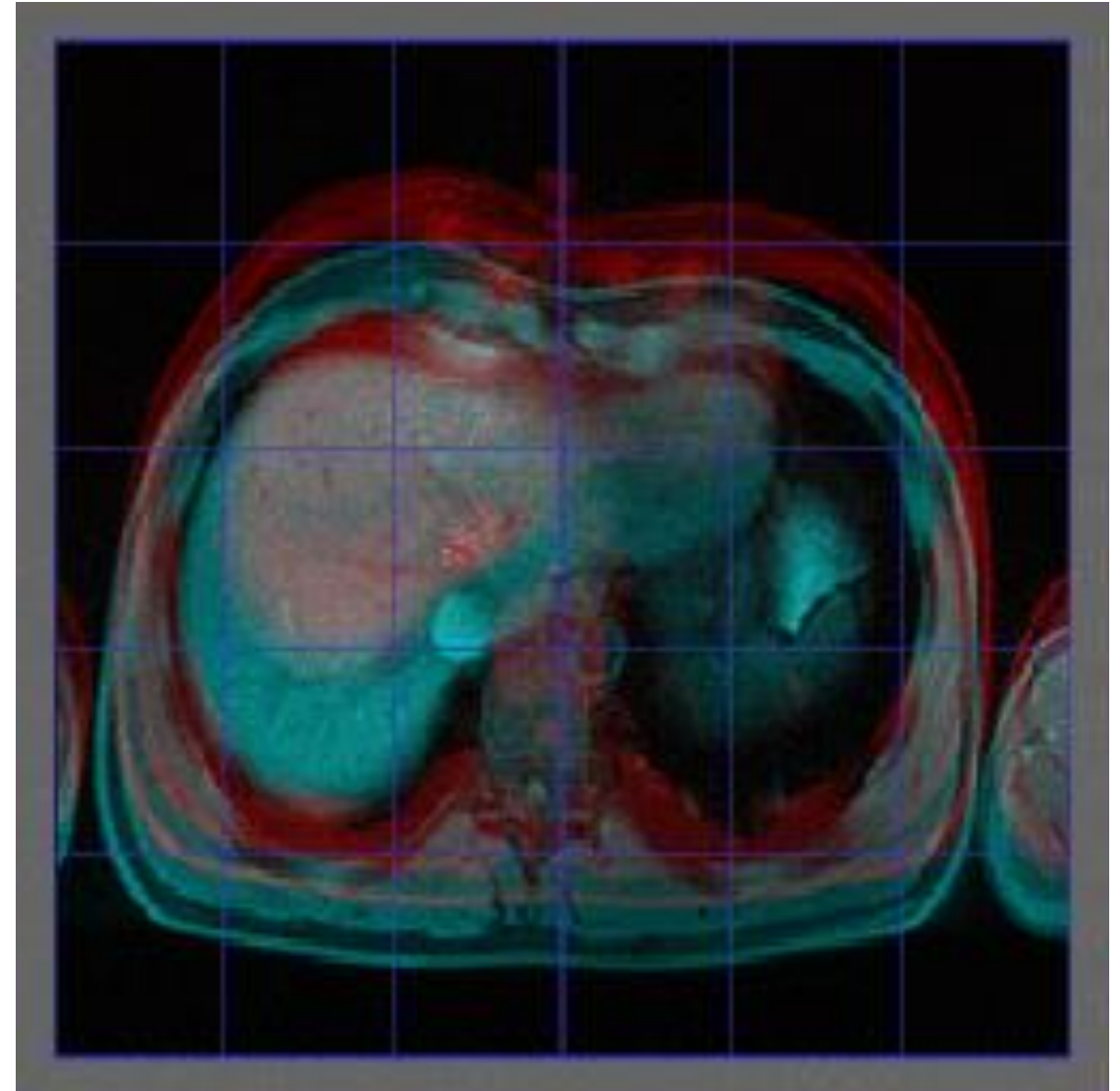
What is Registration?

- Spatial transformation aligning a moving image to a fixed image so anatomical structures overlap
- Transformations range from rigid (translation/rotation) to fully deformable (per-voxel displacement)
- **Classical:** iterative optimisation of a similarity metric (NCC, MI)
 - accurate but slow
- **Deep learning:** a trained network predicts the transformation in a single forward pass

Medical Image Registration

What is Registration?

- Spatial transformation aligning a moving image to a fixed image so anatomical structures overlap
- Transformations range from rigid (translation/rotation) to fully deformable (per-voxel displacement)
- **Classical:** iterative optimisation of a similarity metric (NCC, MI)
 - accurate but slow
- **Deep learning:** a trained network predicts the transformation in a single forward pass



Medical Image Registration

What is Registration?

- Spatial transformation aligning a moving image to a fixed image so anatomical structures overlap
- Transformations range from rigid (translation/rotation) to fully deformable (per-voxel displacement)
- **Classical:** iterative optimisation of a similarity metric (NCC, MI)
 - accurate but slow
- **Deep learning:** a trained network predicts the transformation in a single forward pass

Why Can't We Just Do This Manually?

- Side-by-side slice comparison works for simple in-plane shifts - but deformable motion moves structures **out of plane**
- A slice at position z in the fixed image may correspond to no single slice in the moving image
 - slices never truly match
- Specifying voxel-level correspondences by hand is geometrically infeasible

Uses of Medical Image Registration

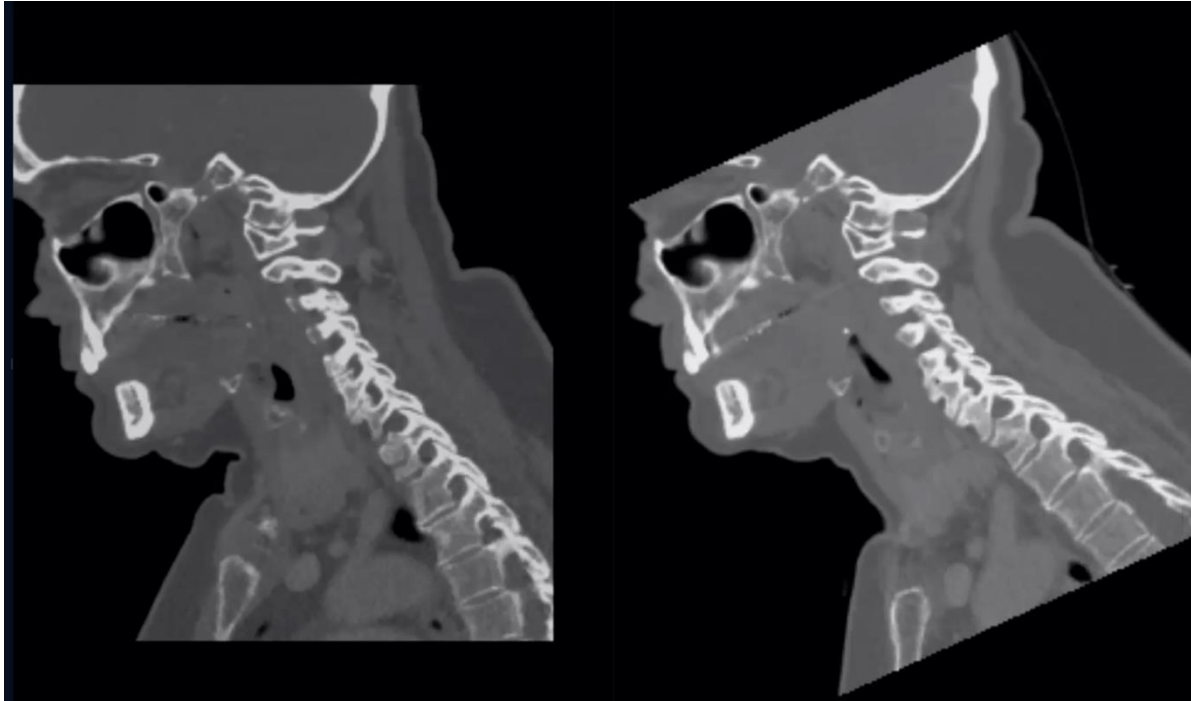
Preprocessing for Downstream Analysis

- Brings images into a common space so other algorithms can operate on them
 - atlas-based segmentation propagation
 - longitudinal change detection (tumour growth, treatment response)
 - multi-modal fusion (PET/CT, MRI/US)
 - radiotherapy dose accumulation

Direct Clinical Use by Radiologists

- Registration result is reviewed, interpreted, and acted on directly
- Should be **visible, trustworthy, and correctable**
 - biopsy targeting (MR-to-TRUS prostate)
 - tumor change assessment
- This is where **visualisation and interactive DL approaches** become useful

Uses of Medical Image Registration



Direct Clinical Use by Radiologists

- Registration result is reviewed, interpreted, and acted on directly
- Should be **visible, trustworthy, and correctable**
 - biopsy targeting (MR-to-TRUS prostate)
 - tumor change assessment
- This is where **visualisation and interactive DL approaches** become useful

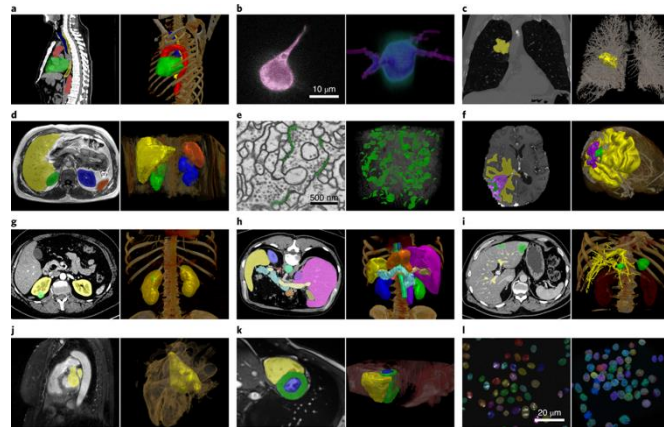
Medical Image Segmentation

Traditional Manual Contouring



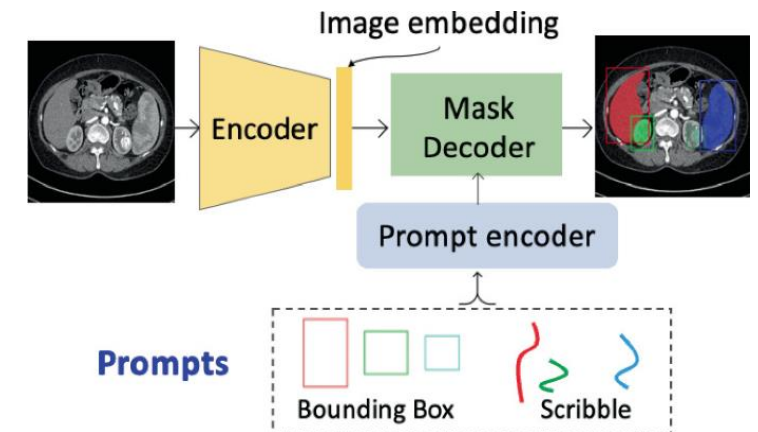
- Highly tedious.
- Time-consuming.
- Prone to massive inter-observer variability.

Task-Specific Experts



- Self-configuring rule sets to handle task-specific hyperparameters automatically.
- Highly rigid.

Promptable Foundation Models



- Trained on millions of diverse medical images.
- Enable zero-shot generalization.
- Interactive clinical workflows.

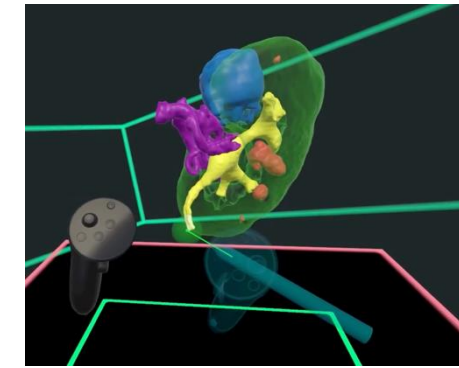
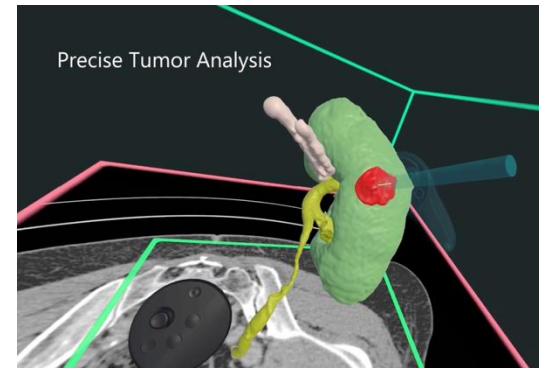
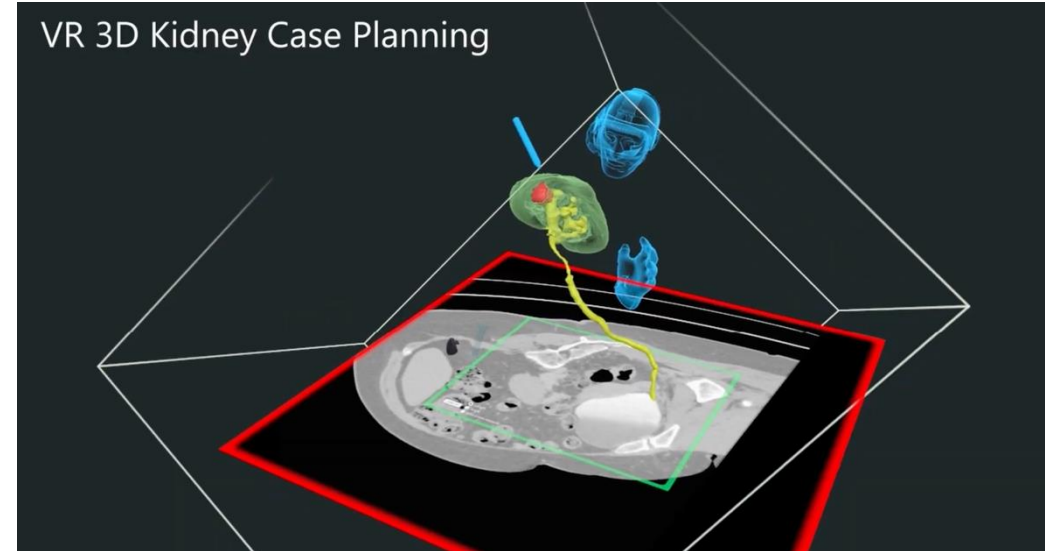
Virtual and Augmented Reality

Advances in Virtual Reality (VR) and head mounted technologies have enabled new approaches to medical image visualization, enabling [1]:

- Realistic high-definition 3D renderings with better depth perception and spatial awareness.
- Intuitive user interactions through hand gestures and haptic feedback.
- Integration with augmented and mixed reality environments.

[1] On the Use of Virtual Reality for Medical Imaging Visualization. Pires et al. Journal of Digital Imaging (2021).

[2] A Pilot Clinical and Technical Validation of an Immersive Virtual Reality Platform for 3D Anatomical Modeling and Contouring in Support of Surgical and Radiation Oncology Treatment Planning. Belec et al. Journal of Informatics in Medicine (2024).

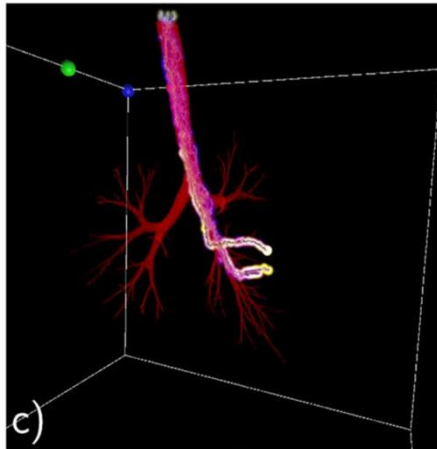
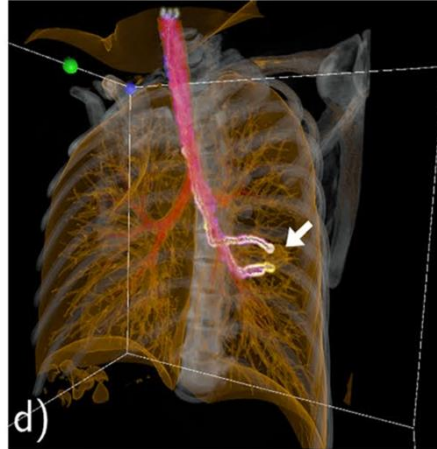


[2]

VR for Medical Imaging

Medical Image Visualization

- Patient-specific anatomy rendering for surgical planning.
- 3D display of medical images (segmentations, multi-modality)



[3]

Simulation of Clinical Environments

- Surgery Simulation
- Clinical Workflow Simulation

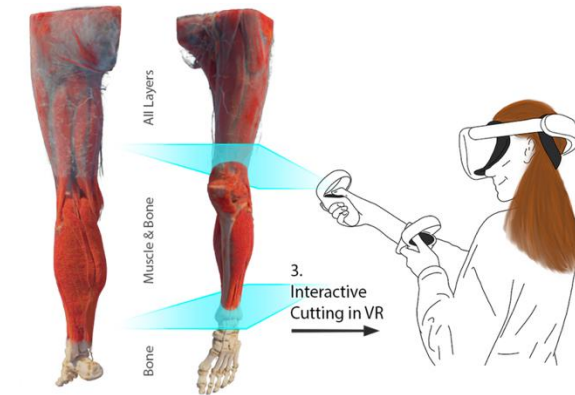
[3] Virtual reality-augmented differentiable simulations for digital twin applications in surgical planning. Cremese et al. Nature (2025)

[4] Multi-Layer Gaussian Splatting for Immersive Anatomy Visualization. Kleinbeck et al. IEEE TRANSACTIONS ON VISUALIZATION AND COMPUTER GRAPHICS. (2025)

[5] Application of 3D Gaussian Splatting for Cinematic Anatomy on Consumer Class Devices. Niedermayr et al. (2024).

Medical training and teaching

- Realistic anatomy rendering
- Replay of medical procedures



[4]



[5]

Papers for Teaching and Presentation

MEDICAL IMAGE REGISTRATION (Papers of your choice are fine too)

- Balakrishnan, Guha, et al. "Voxelmorph: a learning framework for deformable medical image registration." *IEEE transactions on medical imaging* 38.8 (2019): 1788-1800.
- Hoffmann, Malte, et al. "SynthMorph: learning contrast-invariant registration without acquired images." *IEEE transactions on medical imaging* 41.3 (2021): 543-558.
- Tian, Lin, et al. "Gradicon: Approximate diffeomorphisms via gradient inverse consistency." *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*. 2023.
- Demir, Başar, et al. "Multigradicon: A foundation model for multimodal medical image registration." *International Workshop on Biomedical Image Registration*. Cham: Springer Nature Switzerland, 2024.
- Baum, Zachary MC, Yipeng Hu, and Dean C. Barratt. "Meta-learning initializations for interactive medical image registration." *IEEE transactions on medical imaging* 42.3 (2022): 823-833.
- Evan, M. Yu, et al. "Keymorph: Robust multi-modal affine registration via unsupervised keypoint detection." *Medical imaging with deep learning*. 2022.

Papers for Teaching and Presentation

MEDICAL IMAGE SEGMENTATION (Papers of your choice are fine too)

- F. Isensee, P. F. Jaeger, S. A. A. Kohl, J. Petersen, and K. H. Maier-Hein, "nnU-Net: a self-configuring method for deep learning-based biomedical image segmentation," Nat Methods, vol. 18, no. 2, pp. 203–211, Feb. 2021.
- Ma, Jun, et al. "Medsam2: Segment anything in 3d medical images and videos." arXiv preprint arXiv:2504.03600 (2025).
- Rokuss, Maximilian, et al. "Voxtell: Free-text promptable universal 3d medical image segmentation." Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. 2026.
- Ayobi, Nicolás, et al. "Matis: Masked-attention transformers for surgical instrument segmentation." 2023 IEEE 20th International Symposium on Biomedical Imaging (ISBI). IEEE, 2023.
- Isensee, Fabian, et al. "nnInteractive: Redefining 3D promptable segmentation." arXiv preprint arXiv:2503.08373 (2025).
- Ma, Jun, et al. "Medsam2: Segment anything in 3D medical images and videos." arXiv preprint arXiv:2504.03600 (2025).

Papers for Teaching and Presentation

VIRTUAL AND AUGMENTED REALITY (Papers of your choice are fine too)

- A Pilot Clinical and Technical Validation of an Immersive Virtual Reality Platform for 3D Anatomical Modeling and Contouring in Support of Surgical and Radiation Oncology Treatment Planning. Belec et al. Journal of Informatics in Medicine (2024).
- Virtual reality-augmented differentiable simulations for digital twin applications in surgical planning. Cremese, R., et al. Nature Sci Rep 15, 24377 (2025). <https://doi.org/10.1038/s41598-025-99608-2>
- Multi-Layer Gaussian Splatting for Immersive Anatomy Visualization. Kleinbeck et al. IEEE Transactions On Visualization And Computer Graphics. (2025)
- Application of 3D Gaussian Splatting for Cinematic Anatomy on Consumer Class Devices. Niedermayr et al. (2024). Preprint. <https://doi.org/10.48550/arXiv.2404.11285>
- Advanced Medical Use of Three-Dimensional Imaging in Congenital Heart Disease: Augmented Reality, Mixed Reality, Virtual Reality, and Three Dimensional Printing. Goo HW, et al. Korean J Radiol. (2020). doi:10.3348/kjr.2019.0625.
- Virtual Reality to Enhance Understanding of Congenital Heart Disease. Narasimhan SL, et al. Journal of Cardiovascular Development and Disease. (2025). <https://doi.org/10.3390/jcdd12120495>
- Essential Surgical Plan Modifications After Virtual Reality Planning in 50 Consecutive Segmentectomies. Bakhuis W, et al. The Annals of Thoracic Surgery (2022).
- Virtual reality and artificial intelligence for 3-dimensional planning of lung segmentectomies. Sadeghi AH, et al. JTCVS Tech. (2021) doi: 10.1016/j.xjtc.2021.03.016.